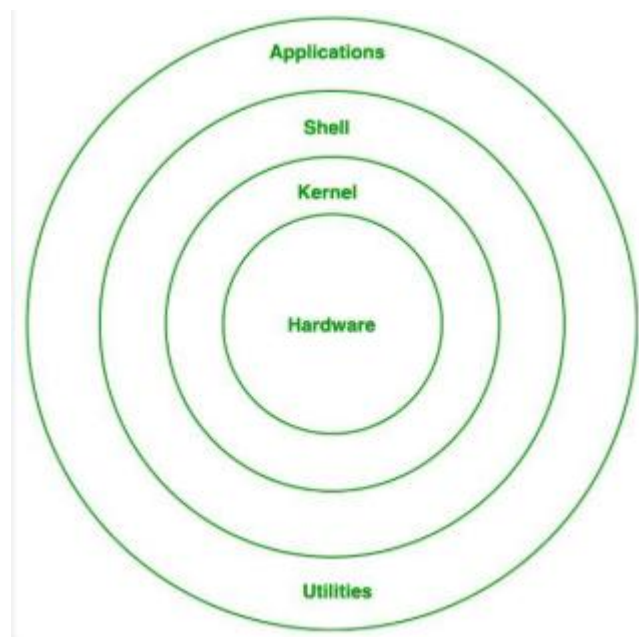


THE LINUX ARCHITECTURE OVERVIEW



Kernel is a monolithic type directly interacts with the hardware. Kernel is the main component which does the interaction with the hardware which is closely tied up with the hardware. So that's the kernel that is surrounded surrounding the hardware.

And Top of the Kernel we have shell which is very similar to egg shell it the shell which is available to user or any applications to interact with it. Through the shell we can give commands to the kernel

- ❖ Kernel is a intermediate between software and hardware components.
- ❖ Kernel provides security and stability to a system

What is Booting process?

The boot process is the sequence of steps a computer goes through when you power it on, starting from hardware checks and firmware initialization, and ending with the operating system being loaded into memory so the system becomes usable.

LINUX BOOT PROCESS

BIOS	Basic Input/Output System executes MBR	❑ BIOS – Does IO check, executes the MBR booting loader
MBR	Master Boot Record executes GRUB	❑ MBR – Important part of the Linux Boot stored in 1 st sector (/dev/hda or /dev/sda) in 512 bytes having ❑ Primary boot loader info in 1st 446 bytes ❑ Partition table information in the next 64 bytes ❑ MBR validation check in the last 2 bytes.
GRUB	Grand Unified Boot Loader executes Kernel	❑ GRUB – Grand Unified Boot loader – loads the OS image(kernel & initrd) (/boot/grub/grub.cfg)
Kernel	Kernel executes /sbin/init	❑ Kernel – Mounts the RFS (/sbin/init) – Initial Ram disk
Init	Init executes runlevel programs	
Runlevel	Runlevel programs are executed from /etc/rc.d/rc*.d/	

INIT

- ❑ Init is the first program that is run by the kernel which will call the other start up scripts like hw, network, GUI etc..
 - It will run the rc system script (/etc/rc.sysinit)
 - Then it will find the default level from file "/etc/inittab" so that it runs the start up script appropriately
 - Then it will run all the linux services that need to be started based on the scripts in from /etc/init.d/ (for eg. ssh start/stop/status/restart etc..)
- ❑ The runlevels in Linux are used to control processes during bootup. Current runlevel of the system can be found by issuing the command #runlevel
- ❑ There are seven(7) run levels available with the Linux operating system:
 - 0 – halt
 - 1 – single-user mode
 - 2 – Multiuser, without NFS
 - 3 – Full multiuser mode
 - 4 – unused
 - 5 – X11
 - 6 – reboot

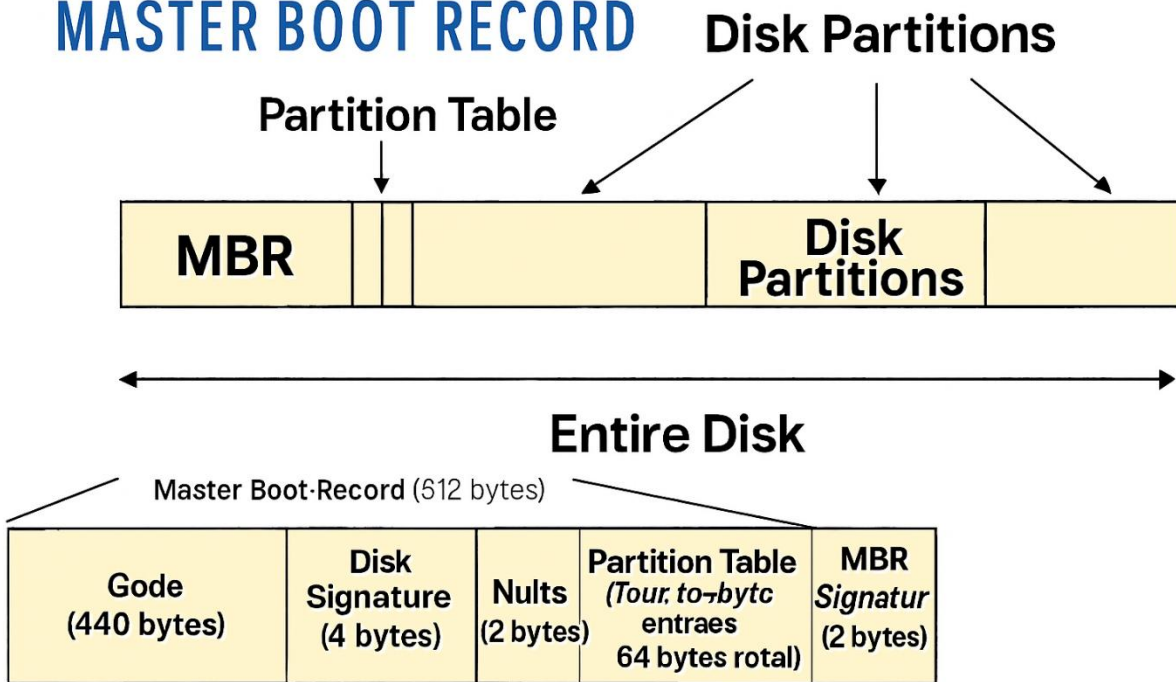
INIT:

Here are the different runlevel

- 0 stands for halt means nothing but the shutdown
- 1 stands for single user mode which means it bring up the linux in single user mode
- 2 stand for bring up the linux in multi user mode which means multiple users cane use

- 3 is full multi user
- 4 is unused
- 5 is the “X 11” mode which is GUI mode for operating system
- 6 is the reboot.

MASTER BOOT RECORD



RUN LEVELS

- ❑ A runlevel can simply be thought of as the state your system enters like if a system
- ❑ A run level is a state of init and the whole system that defines what system services are operating. They are identified by numbers.
- ❑ The system will execute the run level scripts based on the default run level located in the following specific directories:
 - Run level 0 – /etc/rc.d/rc0.d/
 - Run level 1 – /etc/rc.d/rc1.d/
 - Run level 2 – /etc/rc.d/rc2.d/
 - Run level 3 – /etc/rc.d/rc3.d/
 - Run level 4 – /etc/rc.d/rc4.d/
 - Run level 5 – /etc/rc.d/rc5.d/
 - Run level 6 – /etc/rc.d/rc6.d/
- ❑ The programs with name after S or K means which of them to be started or stopped(killed)

To change from one runlevel to another #sudo telinit 3

SHELL START UP

- ❑ Interactive login starts by automatically executing one or more predefined script files
- ❑ Bourne shell (`/bin/bash`) execute

```
$/bin/bash
```

The Bourne shell (`sh`) is a foundational Unix shell developed by Stephen Bourne in 1979, and it remains influential in Linux today as the basis for POSIX-compliant scripting. Most Linux systems use Bash or Dash as modern replacements, but `/bin/sh` often links back to Bourne-compatible behavior.

BASIC BASH COMMANDS

- `ls` — List directory contents
- `echo` — Prints text to the terminal window
- `touch` — Creates a file
- `mkdir` — Create a directory
- `chmod` — Sets the file permissions flag on a file or folder
- `man` — Print manual or get help for a command
- `pwd` — Print working directory
- `cd` — Change directory
- `mv` — Move or rename directory
- `rmdir` — Remove directory
- `Date` — prints the current date